

Name: _____

Date: _____

Free Fall: Reading Motion from Tables

For this worksheet, UP is positive (+) and DOWN is negative (−).

When acceleration is constant (like during free fall), we can find the average velocity over a time interval by averaging the velocities at the beginning and end of the interval.

1. Dropped on Earth

A marble is dropped from rest. On Earth, $v_0 = 0$ m/s and $a = -10$ m/s².

Time t (s)	Velocity v (m/s)	Average velocity since $t = 0$ (m/s)	Position (m)
0	0	–	0
1	–10	–5	–5
2	–20		
3			
4			
5			

1. Complete the table. Start with velocity. Then find average velocity since $t = 0$. Finally, use average velocity $\times$ time to find position.
2. What is the velocity at $t = 3$ s?
3. What is the position at $t = 3$ s?
4. What is the displacement (change in position) from $t = 2$ s to $t = 5$ s?
5. What is the average velocity from $t = 2$ s to $t = 5$ s?

2. Dropped on Another World

A rock is dropped from rest. Here, $v_0 = 0 \text{ m/s}$ and $a = -4 \text{ m/s}^2$.

Time t (s)	Velocity v (m/s)	Average velocity since $t = 0$ (m/s)	Position (m)
0	0	–	0
1	–4		
2			
3			
4			
5			
6			

1. Complete the table.
2. How much does velocity change each second?
3. Find the displacement from $t = 2 \text{ s}$ to $t = 5 \text{ s}$.
4. Find the average velocity from $t = 2 \text{ s}$ to $t = 5 \text{ s}$.
5. Does the rock move the same distance during every one-second interval? Use numbers from the table to support your answer.

3. Thrown Downward on Earth

A ball is thrown downward at 5 m/s. Thus, $v_0 = -5$ m/s and $a = -10$ m/s².

Time t (s)	Velocity v (m/s)	Average velocity since $t = 0$ (m/s)	Position (m)
0	-5	-	0
1	-15		
2			
3			
4			
5			

1. Complete the table.
2. What is the velocity at $t = 4$ s?
3. Find the displacement from $t = 1$ s to $t = 4$ s.
4. Find the average velocity from $t = 1$ s to $t = 4$ s.
5. Compare this velocity table with a dropped object on Earth. What stays the same from one row to the next? What is different at $t = 0$?

4. Thrown Downward on Another World

A stone is thrown downward at 6 m/s. Here, $v_0 = -6$ m/s and $a = -4$ m/s².

Time t (s)	Velocity v (m/s)	Average velocity since $t = 0$ (m/s)	Position (m)
0	-6	-	0
1			
2			
3			
4			
5			
6			

1. Complete the table.

2. What is the velocity at $t = 5$ s?

3. What is the position at $t = 5$ s?

4. Find the displacement from $t = 2$ s to $t = 6$ s.

5. Find the average velocity from $t = 2$ s to $t = 6$ s.

6. At $t = 4$ s, write the instantaneous velocity and the average velocity since $t = 0$.

5. Thrown Downward: Less Help

A ball is thrown downward at 8 m/s where $a = -6 \text{ m/s}^2$. Thus, $v_0 = -8 \text{ m/s}$.

Time t (s)	Velocity v (m/s)	Average velocity since $t = 0$ (m/s)	Position (m)
0	-8	-	0
1			
2			
3			
4			
5			
6			

1. Complete the table.
2. Find the displacement from $t = 1 \text{ s}$ to $t = 5 \text{ s}$.
3. Find the average velocity from $t = 1 \text{ s}$ to $t = 5 \text{ s}$.
4. At $t = 6 \text{ s}$, which is more negative: the instantaneous velocity or the average velocity since $t = 0$?
5. How can you tell from the table that the ball is speeding up as it moves downward?

6. Thrown Upward on Earth

A ball is thrown upward at 20 m/s. Thus, $v_0 = +20$ m/s while $a = -10$ m/s².

Time t (s)	Velocity v (m/s)	Average velocity since $t = 0$ (m/s)	Position (m)
0	20	–	0
1	10		
2			
3			
4			
5			

1. Complete the table.
2. At what time is the velocity 0 m/s?
3. What is the position at that moment?
4. What happens to the direction of motion when velocity changes from positive to negative?
5. When does the ball return to its starting position?
6. What is its average velocity from $t = 0$ until when the ball returns to its starting position?

7. Thrown Upward on Another World

A ball is thrown upward at 12 m/s on a world where $a = -4 \text{ m/s}^2$.

Time t (s)	Velocity v (m/s)	Average velocity since $t = 0$ (m/s)	Position (m)
0	12	–	0
1			
2			
3			
4			
5			
6			

1. Complete the table.
2. When does the ball reach its highest position?
3. What is the highest position reached?
4. Find the displacement from $t = 1 \text{ s}$ to $t = 5 \text{ s}$.
5. Find the average velocity from $t = 1 \text{ s}$ to $t = 5 \text{ s}$.
6. During which times is the ball moving upward? During which times is it moving downward?

8. Put It Together

A small object starts moving upward at 8 m/s. The planet's gravitational acceleration is -2 m/s^2 .

Write the starting values: $v_0 =$ _____ m/s, $a =$ _____ m/s^2 .

Time t (s)	Velocity v (m/s)	Average velocity since $t = 0$ (m/s)	Position (m)
0	8	–	0
1			
2			
3			
4			
5			
6			
7			

1. Complete the table.
2. When does the object change direction?
3. What is its position when it changes direction?
4. Find the displacement from $t = 2 \text{ s}$ to $t = 7 \text{ s}$.
5. Find the average velocity from $t = 2 \text{ s}$ to $t = 7 \text{ s}$.
6. At $t = 7 \text{ s}$, is the object moving up or down? What table entry tells you?